

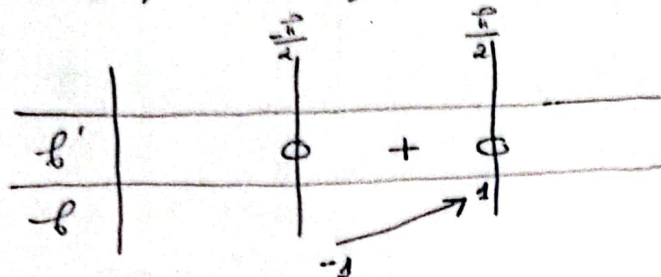
* fct sinus

fonction usuelle

$$\mathcal{C} : \left[-\frac{\pi}{2}; \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

$$x \mapsto \sin x$$

$$\mathcal{C}'(x) = \cos(x) = 0 \iff x = \left\{ -\frac{\pi}{2}; \frac{\pi}{2} \right\}$$



f est une fct stricte $\nearrow$ sur $\left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$

f admet une fct réciproque de $\mathcal{C}$ sur $\mathcal{C}\left(\left[-\frac{\pi}{2}; \frac{\pi}{2}\right]\right) = [-1, 1]$

$\mathcal{C}^{-1}$, notée **arc sin x**

soit $x \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]; y \in [-1, 1]$ tq **$\sin x = y$**

$$\Rightarrow \mathbf{x = \arcsin(y)}$$

• Rq: $\sin(\arcsin y) = y; \forall y \in [-1, 1]$

* $\arcsin(\sin x) = x; \forall x \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$

* $\arcsin\left(\sin \frac{12\pi}{7}\right)$ on a $\frac{12\pi}{7} = -\frac{2\pi}{7}$

donc $\sin\left(\frac{12\pi}{7}\right) = \sin\left(-\frac{2\pi}{7}\right)$

or $-\frac{2\pi}{7} \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$

$$\arcsin\left(\sin\left(\frac{12\pi}{7}\right)\right) = \arcsin\left(-\frac{2\pi}{7}\right) = -\frac{2\pi}{7}$$

* $\sin^2 x + \cos^2 x = 1 \implies \cos x = \pm \sqrt{1 - \sin^2 x}$

• $\cos(\arcsin y); y \in [-1, 1] \left\{ \begin{array}{l} \cos(\arcsin y) = \sqrt{1 - y^2} \\ = +\sqrt{1 - \sin^2(\arcsin y)} = \sqrt{1 - y^2} \end{array} \right.$

• $\text{Tg}(\arcsin y) = \frac{\sin(\arcsin y)}{\cos(\arcsin y)} = \frac{y}{\sqrt{1 - y^2}}; \forall y \in]-1, 1[$

$$\mathbf{\text{Tg}(\arcsin y) = \frac{y}{\sqrt{1 - y^2}}}$$

Derivé de arc sin :

arc sin y est dérivable sur $[-1, 1]$

1) domaine de dérivabilité de arc sin

$$f'(x) = 0 \implies x = x_1$$

$$f(x_1) = 0$$

f^{-1} n'est pas d en $f(x_1)$

Pour ce cas on a

$$f(x) = \sin x \implies f'(x) = \cos x, f'(x) = 0 \implies x = \left\{ \pm \frac{\pi}{2} \right\}$$

$$f\left(-\frac{\pi}{2}\right) = -1 ; f\left(\frac{\pi}{2}\right) = 1 \text{ donc } f^{-1} \text{ n'est pas d en } \{\pm 1\}$$

$$D(f^{-1})' =]-1, 1[$$

2) Calcul du arc sin'(y) ; $(f^{-1})'$.

$$\text{soit } f(x) = y \implies \sin x = y \implies x = \arcsin(y)$$

$$x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\forall y \in]-1, 1[$$

$$y \in [-1, 1].$$

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{1}{\cos(\arcsin(y))} = \frac{1}{\sqrt{1-y^2}}$$

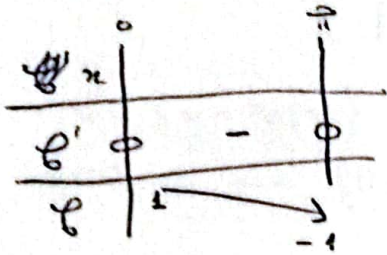
$f^{ct} \cos$

$$\cos(-\alpha) = \cos(\alpha)$$

Soit $f: [0, \pi] \rightarrow [-1, 1]$

$$x \mapsto \cos x$$

$$f'(x) = -\sin x; f'(x) = 0 \Rightarrow x = \{0, \pi\}$$



* f est $\leq$ et strictement $\searrow$ sur $[0, \pi]$
du fait qu'il admet une f^{ct} réciproque
de $f([0, \pi]) = [-1, 1]$ notée $\arccos$.

$$\text{soit } \cos x = y; x \in [0, \pi]; y \in [-1, 1]$$

$$x = \arccos(y)$$

$$\text{Rq: } \cos(\arccos(y)) = y, \forall y \in [-1, 1]$$

$$\arccos(\cos x) = x, \forall x \in [0, \pi]$$

exercice:

$$\arccos\left(\cos\left(\frac{17\pi}{5}\right)\right)$$

$$\frac{17\pi}{5} = \frac{20\pi}{5} - \frac{3\pi}{5}; \cos\left(\frac{17\pi}{5}\right) = \cos\left(-\frac{3\pi}{5}\right)$$

$$\text{or } \cos(-\alpha) = \cos \alpha \text{ donc } \cos\left(-\frac{3\pi}{5}\right) = \cos\left(\frac{3\pi}{5}\right)$$

$$\text{et } \frac{3\pi}{5} \in [0, \pi]$$

$$\arccos\left(\cos\left(\frac{17\pi}{5}\right)\right) = \arccos\left(\cos\left(\frac{3\pi}{5}\right)\right) = \frac{3\pi}{5}$$

$$\star \cos(\arcsin(y))$$

$$1 + \sqrt{1 - \sin^2(\arcsin(y))}$$

$$= \sqrt{1 - y^2} \quad \forall y \in [-1, 1]$$

$$\star \sin(2 \arcsin(y)) \text{ ou } \sin(2\alpha) = 2 \sin \alpha \cos \alpha$$

$$= 2 \sin(\arcsin(y)) \cdot \cos(\arcsin(y)) = 2y\sqrt{1-y^2}$$

⊙ domaine de dérivabilité de $\arccos y$

$$f = \cos x, f'(x) = 0 \Rightarrow x = \{0, \pi\}$$

$$f(0) = 1; f(\pi) = -1$$

donc $\arccos$ n'est pas d en (± 1)

$\Rightarrow$ domaine de dérivabilité de $\arccos$ est $] -1, 1[$.

calcul de $\arccos'(y)$

$$\text{soit } f(x) = y \Leftrightarrow \cos x = y \quad x \in [0, \pi]$$

$$y \in [-1, 1]$$

$$x = \arccos(y)$$

$$\arccos'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{-1}{\sin(f^{-1}(y))}$$

$$= \frac{-1}{\sin(\arccos(y))} = \frac{-1}{\sqrt{1-y^2}} \quad \forall y \in]-1, 1[$$

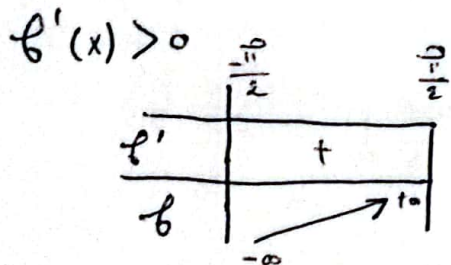
* f^{ct} tangente

$$\text{tg}(x) = \frac{\sin(x)}{\cos(x)}$$

$$\text{soit } f :]-\frac{\pi}{2}; \frac{\pi}{2}[\rightarrow \mathbb{R}$$

$$x \mapsto \text{tg}(x)$$

$$f(x) = \frac{\sin(x)}{\cos(x)} ; f'(x) = \frac{\cos x \cdot \cos x - (-\sin x) \cdot \sin x}{\cos^2(x)}$$
$$= \frac{1}{\cos^2 x} = 1 + \text{tg}^2(x).$$



• f est continue et strictement $\nearrow$ sur $] -\frac{\pi}{2}; \frac{\pi}{2}[$ donc f réalise une bijection de $] -\frac{\pi}{2}; \frac{\pi}{2}[$ sur $f(]-\frac{\pi}{2}; \frac{\pi}{2}[) = \mathbb{R}$.

f admet une f^{ct} réciproque de f sur $\mathbb{R}$ notée $\arctg$.

$$\bullet \text{tg}(x) = y ; x \in]-\frac{\pi}{2}; \frac{\pi}{2}[; y \in \mathbb{R}$$

donc $x = \arctg(y)$.

$$\bullet \text{tg}(\arctg(y)) = y, \quad \forall y \in \mathbb{R}.$$

$$\bullet \arctg(\text{tg}(x)) = x ; \forall x \in]-\frac{\pi}{2}; \frac{\pi}{2}[$$

$$\bullet \cos(\arctg(y))$$

On a $\cos^2 x = \frac{1}{1 + \tan^2(x)}$

$$\cos x = \pm \frac{1}{\sqrt{1 + \tan^2(x)}}$$

donc $\cos(\arctan(y)) = \frac{1}{\sqrt{1 + \tan^2(\arctan(y))}} = \frac{1}{\sqrt{1 + y^2}}$

• $\sin(\arctan(y))$

On a $\sin x = \sqrt{1 - \cos^2(\arctan(y))}$

$$= \sqrt{1 - \frac{1}{1 + y^2}} = \sqrt{\frac{y^2}{1 + y^2}} = \frac{|y|}{\sqrt{1 + y^2}}$$

Derivée de $\arctan$:

• $f'(x) = 0 \Leftrightarrow 1 + \tan^2(x) = 0$

or $1 + \tan^2(x) \neq 0 \quad \forall x \Rightarrow f^{-1}$ est $\downarrow$ sur $\mathbb{R}$.

• soit $f(x) = y \quad ; x \in]-\frac{\pi}{2}; \frac{\pi}{2}[\quad ; y \in \mathbb{R}$.

c'est à dire $\tan(x) = y$

$$x = \arctan(y) = f^{-1}(y)$$

• $\forall y \in \mathbb{R}$

$$(f^{-1})'(y) = \arctan'(y) = \frac{1}{f'(f^{-1}(y))}$$

$$= \frac{1}{1 + \tan^2(\arctan(y))} = \frac{1}{1 + y^2}$$

Exercice :

$$f(x) = \frac{1}{\sqrt{\cos x}} ; x \in [0, \frac{\pi}{2}[$$

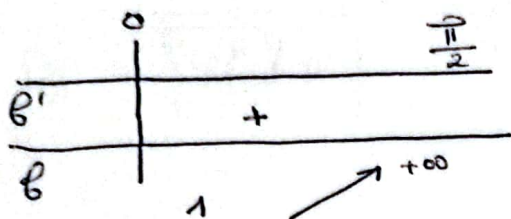
1) étudier f

2) calculer $(f^{-1})'$

a) $f(x) = \frac{1}{(\cos(x))^{\frac{1}{2}}}$

$$f'(x) = -\frac{1}{2} \cdot \frac{-\sin x}{(\cos x)^{\frac{3}{2}}} = \frac{\sin x}{2 \cdot \cos x \sqrt{\cos x}} = f'(x)$$

$$f'(x) = 0 \Rightarrow \sin x = 0 \Rightarrow x = 0$$



f est cont strict $\uparrow$ sur $[0, \frac{\pi}{2}[$ donc f admet une f^{ct} réciproque sur $f([0, \frac{\pi}{2}[) = [1; +\infty[$ notée f^{-1}

• soit $f(x) = y$

$$\frac{1}{\sqrt{\cos x}} = y$$

$$\sqrt{\cos x} = \frac{1}{y}$$

$$\cos x = \left(\frac{1}{y}\right)^2$$

$$x = \arccos \frac{1}{y^2}$$

$$f^{-1} = \arccos\left(\frac{1}{y^2}\right)$$

• Domaine de dérivabilité de f^{-1}

$$\text{soit } f'(x) = 0 \Rightarrow x = 0 \Rightarrow f(0) = 1$$

f^{-1} est d sur $[1; +\infty[$ donc sur $]1; +\infty[$

2) $(f^{-1})'$: on a $f(x) = y$

$$(f^{-1})(y) = \frac{1}{f'(f^{-1}(y))} = \frac{2 \cdot \cos(f^{-1}(y))^{\frac{3}{2}}}{\sin(f^{-1}(y))}$$

$$\cos(\varphi^{-1}(y)) = \cos\left(\arccos\left(\frac{1}{y^2}\right)\right) = \frac{1}{y^2}$$

$$\bullet \sin(\varphi^{-1}(y)) = \sin\left(\arccos\left(\frac{1}{y^2}\right)\right) = \sqrt{1 - \left(\frac{1}{y^2}\right)^2}$$

$$\boxed{\sin(\arccos(y)) = \sqrt{1 - \cos^2(\arccos(y))} = \sqrt{1 - y^2}} \quad \text{This}$$

$$(\varphi^{-1}(y))' = \frac{2 \cdot \left(\frac{1}{y^2}\right)^{\frac{3}{2}}}{\sqrt{1 - \left(\frac{1}{y^2}\right)^2}} = \frac{2}{y^3 \cdot \frac{\sqrt{y^4 - 1}}{y^2}} = \frac{2}{y\sqrt{y^4 - 1}}$$

Exercice

$$\text{soit } g(x) = \arcsin\left(\frac{x}{\sqrt{x^2 + 1}}\right)$$

$$Dg = \left\{ x \in \mathbb{R} \mid -1 \leq \frac{x}{\sqrt{x^2 + 1}} \leq 1 \right\}$$

$$\bullet \frac{x}{\sqrt{x^2 + 1}} - 1 = \frac{x - \sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} = \frac{x^2 - (x^2 + 1)}{(x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1})}$$

$$= \frac{-1}{(x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1})} < 0$$

$$\frac{x}{\sqrt{x^2 + 1}} < 0$$

$$\bullet -1 - \frac{x}{\sqrt{x^2 + 1}} = \frac{-\sqrt{x^2 + 1} - x}{\sqrt{x^2 + 1}} = \frac{-1}{(\sqrt{x^2 + 1} - x)\sqrt{x^2 + 1}} < 0$$

$$\text{et } -1 - \frac{x}{\sqrt{x^2 + 1}} < 0 \Rightarrow -1 < \frac{x}{\sqrt{x^2 + 1}}$$

$$\forall x \text{ on a } -1 \leq \frac{x}{\sqrt{x^2 + 1}} \leq 1 \quad Dg = \mathbb{R}$$

$$\arcsin(g(x)) \text{ est def sur } -1 \leq g(x) \leq 1$$

$$\frac{d}{dx} \arcsin(g(x)) = g'(x) \cdot \frac{1}{\sqrt{1 - g(x)^2}}$$

$$\arcsin'(x) = \frac{1}{\sqrt{1 - x^2}} ; -1 < x < 1$$

$$* f(x) = \underbrace{\arcsin}_{h} \left(\underbrace{\frac{x}{\sqrt{x^2+1}}}_{g} \right)$$

$$D_f = \mathbb{R}.$$

$$f'(x) = (h \circ g(x))' \\ = g'(x) \cdot h'(g(x))$$

$$g(x) = \frac{x}{\sqrt{x^2+1}}$$

$$g'(x) = \frac{\sqrt{x^2+1} - \frac{2 \cdot x \cdot x}{2\sqrt{x^2+1}}}{x^2+1} = \frac{x^2+1-x^2}{\sqrt{x^2+1} \cdot (x^2+1)} \\ = \frac{1}{\sqrt{x^2+1} \cdot (x^2+1)}$$

$$h(x) = \arcsin(x)$$

$$h'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f'(x) = \frac{1}{\sqrt{1+x^2} \cdot (1+x^2)} \cdot \frac{1}{\sqrt{1-g^2(x)}} \\ = \frac{1}{\sqrt{1+x^2} (1+x^2)} \cdot \frac{1}{\sqrt{1-\frac{x^2}{x^2+1}}} \\ = \frac{1}{\sqrt{1+x^2} \cdot (1+x^2)} \cdot \frac{\sqrt{x^2+1}}{1} \\ = \frac{1}{1+x^2} = f'(x).$$

simplifier:

$$\text{on a } f'(x) = \frac{1}{1+x^2}$$

$$\text{or on sait que } \arctan'(x) = \frac{1}{1+x^2}$$

càd

$$f'(x) = \arctan'(x)$$

$$\text{Pg : si } f'_1 = f'_2 \Rightarrow f_1 = f_2 + \text{cst}$$

$$\left. \begin{aligned} f(n) &= \arctg(n) + \text{cte} \\ f(0) &= \arctg(0) + \text{cte} \\ 0 &= 0 + \text{cte} \Rightarrow \text{cte} = 0 \end{aligned} \right\} f(n) = \arctg(n).$$

Exercice

$$f(n) = \arcsin(n) + \arccos(n) \quad n \in [0, 1]$$

$$f'(n) = \frac{1}{\sqrt{1-n^2}} - \frac{1}{\sqrt{1-n^2}} = 0$$

$$f = \text{cte}$$

$$\begin{aligned} f(0) &= \arcsin(0) + \arccos(0) \\ &= 0 + \frac{\pi}{2} = \frac{\pi}{2} \end{aligned}$$

$$f(n) = \frac{\pi}{2} \quad \forall n \in [0, 1]$$

soit $f(n) = \arctg(n) + \arctg\left(\frac{1}{n}\right)$; $\forall n \in \mathbb{R}^*$

1) calculer f' puis en déduire

$$f(n) = \arctg(n) + \arctg\left(\frac{1}{n}\right)$$

$$f'(n) = \frac{1}{1+n^2} + \left(-\frac{1}{n^2}\right) \cdot \frac{1}{1+\left(\frac{1}{n}\right)^2}$$

$$= \frac{1}{1+n^2} - \frac{1}{n^2 \left[\frac{n^2+1}{n^2} \right]}$$

$$= \frac{1}{1+n^2} - \frac{1}{n^2+1} = 0$$

$$f(n) = \text{cte}$$

$$\begin{aligned} f(1) &= \arctg(1) + \arctg\left(\frac{1}{1}\right) \\ &= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \end{aligned}$$

$$f(n) = \frac{\pi}{2} \quad \text{si } n > 0$$

$$\text{or } f(-n) = \arctg(-n) + \arctg\left(-\frac{1}{n}\right)$$

Exercice: simplifier en précisant chaque fois le D_f :

①: $\cos(\operatorname{arctg}(n))$

$$\cos^2(\operatorname{arctg}(n)) = \frac{1}{1 + \operatorname{tg}^2(\operatorname{arctg}(n))} = \frac{1}{1 + n^2}$$

$$\cos n = \pm \sqrt{\frac{1}{1 + n^2}}$$

② $\sin(\operatorname{arctg}(n))$

$$\begin{aligned} \sin^2(\operatorname{arctg}(n)) &= 1 - \cos^2(\operatorname{arctg}(n)) \\ &= 1 - \frac{1}{1 + n^2} = \frac{n^2}{1 + n^2} \Rightarrow \sin(\operatorname{arctg}(n)) = \frac{|n|}{\sqrt{1 + n^2}} \end{aligned}$$

③ $\operatorname{tg}(2n) = \frac{2 \operatorname{tg}(n)}{1 - \operatorname{tg}^2(n)}$

$$\operatorname{tg}(2 \operatorname{arctg}(n)) = \frac{2 \operatorname{tg}(\operatorname{arctg}(n))}{1 - \operatorname{tg}^2(\operatorname{arctg}(n))} = \frac{2n}{1 - n^2}$$

$$2 \operatorname{arctg}(n) \neq \pm \frac{\pi}{2}$$

$$\operatorname{arctg}(n) \neq \pm \frac{\pi}{4} \Rightarrow n \neq \pm 1 \quad D_f = \mathbb{R} \setminus \{\pm 1\}$$

④ $\cos(4 \operatorname{arctg}(n))$ ou $\cos(2n) = \frac{1 - \operatorname{tg}^2(n)}{1 + \operatorname{tg}^2(n)}$

$$= \cos(2 \cdot (2 \operatorname{arctg}(n)))$$

$$= \frac{1 - \operatorname{tg}^2(2 \operatorname{arctg}(n))}{1 + \operatorname{tg}^2(2 \operatorname{arctg}(n))}$$

$$= \frac{1 - \left(\frac{2n}{1-n^2}\right)^2}{1 + \left(\frac{2n}{1-n^2}\right)^2} = \frac{1 + n^4 - 6n^2}{(1 + n^2)^2}$$

⑤ $\operatorname{tg}(\operatorname{arcsin} n)$

$$D_f =]-1, 1[$$

$$= \frac{\sin(\operatorname{arcsin} n)}{\cos(\operatorname{arcsin} n)} = \frac{n}{\sqrt{1 - n^2}}$$

⑥ $\operatorname{tg}(\arccos(n)) = \frac{\sin(\arccos(n))}{\cos(\arccos(n))} = \frac{\sqrt{1 - n^2}}{n}$

$\arccos n$ est def sur $[-1, 1]$ de plus $\operatorname{tg}(x)$ def $x \neq \pm \frac{\pi}{2}$

Sci $\arccos(n) \neq \pm \frac{\pi}{2} \Rightarrow n \neq 0 \Rightarrow D_f = [-1, 1] \setminus \{0\}$

⑧ $f(x) = \arcsin\left(\frac{2x}{1+x^2}\right)$ exercice complet

$$D_f = \left\{ x \in \mathbb{R} \mid -1 \leq \frac{2x}{1+x^2} \leq 1 \right\}$$

$$\bullet -1 \leq \frac{2x}{1+x^2}$$

$$-1 - \frac{2x}{1+x^2} \leq 0$$

$$\frac{-1 - x^2 - 2x}{1+x^2} \leq 0$$

$$\frac{-(x+1)^2}{1+x^2} \leq 0 \quad \underline{\text{tjv}}$$

$$\bullet \frac{2x}{1+x^2} \leq 1$$

$$\frac{2x}{1+x^2} - 1 \leq 0$$

$$\frac{2x - 1 - x^2}{1+x^2} \leq 0$$

$$\frac{-(x-1)^2}{1+x^2} \leq 0 \quad \underline{\text{tjv}}$$

$$D_f = \mathbb{R} :$$

$$\frac{2x}{1+x^2} = 1 \Rightarrow x = 1$$

$$\frac{2x}{1+x^2} = -1 \Rightarrow x = -1$$

* f n'est pas dérivable en $x = \pm 1$ $\forall x \in \mathbb{R} \setminus \{\pm 1\}$

$$\bullet f'(x) = \frac{\arcsin}{f} \left(\frac{2x}{1+x^2} \right)$$

$$= \left(\frac{2x}{1+x^2} \right) \cdot \frac{1}{\sqrt{1 - \left(\frac{2x}{1+x^2} \right)^2}}$$

$$= \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} \cdot \frac{1}{\sqrt{\frac{(1-x^2)^2}{(1+x^2)^2}}}$$

$$= \frac{2(1-x^2)}{(1+x^2)^2} \cdot \frac{|1+x^2|}{|1-x^2|}$$

$$= \frac{2(1-x^2)}{(1+x^2)(1-x^2)}$$

$$= \begin{cases} \frac{2}{1+x^2} & \text{si } 1-x^2 > 0 \\ \frac{-2}{1+x^2} & \text{si } 1-x^2 < 0 \end{cases}$$

donc $f(x) = \arcsin\left(\frac{2x}{1+x^2}\right) \Rightarrow D_f = \mathbb{R}.$

$$f'(x) = \begin{cases} \frac{2}{1+x^2} & \text{si } 1-x^2 > 0 \\ -\frac{2}{1+x^2} & \text{si } 1-x^2 < 0 \end{cases}$$

1) si $1-x^2 > 0 \Rightarrow x \in]-1, 1[$

$$f'(x) = \frac{2}{1+x^2} = 2 \cdot \frac{1}{1+x^2} = \cancel{\frac{1}{1+x^2}} (2 \cdot \arctan(x))'$$

$f(x) = 2 \arctan(x) + \text{constante} ; \forall x \in]-1, 1[.$

$f(0) = 2 \arctan(0) + \text{constante} \Rightarrow 0 = 2 \cdot 0 + \text{constante}$
 $\text{constante} = 0.$

* si $1-x^2 < 0 \Rightarrow x \in]-\infty, -1[\cup]1, +\infty[$

$$f'(x) = \frac{-2}{1+x^2} = -2 \cdot \frac{1}{1+x^2} = (-2 \arctan(x))'$$

$f(x) = -2 \arctan(x) + \text{constante}$

a) si $x \in]-\infty, -1[; f(x) = -2 \arctan(x) + \text{constante}$

pour $x = -\sqrt{3}$

$f(-\sqrt{3}) = -2 \arctan(\sqrt{3}) + \text{constante}$

$\tan\left(-\frac{\pi}{3}\right) = -\sqrt{3}$

$= -2 \left(-\frac{\pi}{3}\right) + \text{constante}$

$\Rightarrow \arctan(-\sqrt{3}) = -\frac{\pi}{3}$

$= \frac{2\pi}{3} + \text{constante}.$

$\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$

$\Rightarrow \arcsin\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$

$f(-\sqrt{3}) = \arcsin\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$

$\Rightarrow \frac{2\pi}{3} + \text{constante} = -\frac{\pi}{3}$

$\Rightarrow \text{constante} = -\pi$

Conclusion: si $x \in]-\infty, -1[$

$f(x) = -2 \arctan(x) - \pi$

b) si $x \in]1, +\infty[; \text{or } f(x) = -2 \arctan(x) + \text{constante}$

pour $x = \sqrt{3} ; f(\sqrt{3}) = -2 \arctan(\sqrt{3}) + \text{constante}$

donc $\text{constante} = f(\sqrt{3}) + 2 \arctan(\sqrt{3}) = \frac{\pi}{3} + 2 \cdot \frac{\pi}{3} = \pi$

$$f(x) = \begin{cases} -2 \arctan(x) - \pi & \text{si } x \in]-\infty; -1[\\ 2 \arctan(x) & \text{si } x \in]-1, 1[\\ -2 \arctan(x) + \pi & \text{si } x \in]1; +\infty[\end{cases}$$

$$\bullet \lim_{(-1)^-} f(x) = -2 \arctan(-1) - \pi = -\frac{\pi}{2}$$

$$\bullet \lim_{(-1)^+} f(x) = 2 \arctan(-1) = -\frac{\pi}{2}$$

$$\text{or } f(-1) = \arcsin\left(\frac{2 \cdot (-1)}{1 + (-1)^2}\right) = \arcsin(-1) = -\frac{\pi}{2}$$

donc on peut les] [en (-1)

$$\bullet \lim_{1^-} f(x) = 2 \arctan(1) = 2 \frac{\pi}{4} = \frac{\pi}{2}$$

$$\bullet \lim_{1^+} f(x) = 2 \arctan(1) + \pi = 2 \frac{\pi}{4} + \pi = \frac{3\pi}{2}$$

$$\text{et on a } f(1) = \arcsin\left(\frac{2 \cdot 1}{1 + (1)^2}\right) = \arcsin(1) = \frac{\pi}{2}$$

donc on peut faire les] [en 1

conclusion:

$$f(x) = \begin{cases} -2 \arctan(x) - \pi & \text{si } x \in]-\infty, -1[\\ 2 \arctan(x) & \text{si } x \in [-1, 1] \\ -2 \arctan(x) + \pi & \text{si } x \in]1, +\infty[\end{cases}$$